METRO AG SCI Project: Scope Derivation — Frameworks Applied

This document traces how the scope statement, KITs, and KIQs of the METRO AG SCI project were derived from the course frameworks. Each section applies one framework from the SCI lecture material to the METRO case, demonstrating that the choices made are the result of a structured analytical path.

1. Rumsfeld Matrix — surfacing knowns and unknowns

The Rumsfeld Matrix distinguishes four types of knowledge at the start of a project: what we know we know (assumptions), what we know we don't know (gaps), what we don't know we know (tacit knowledge), and what we don't know we don't know (discoveries). Applied to METRO at project start (t = 0):

	…that we know	…that we don't know
What we know…	**Assumptions.** METRO is a European B2B wholesaler serving HoReCa; digital channels (e-commerce platform, mobile commerce app) are publicly documented; the sCore strategy explicitly targets digital sales growth to 40% by 2030.	**Gaps.** How effectively do these digital channels actually retain customers? Which emerging technologies will most reshape HoReCa ordering in the next 3–5 years? How do competitors compare in feature coverage and customer perception?
What we don't know…	**Tacit knowledge.** The broader academic discourse on digital retention, the patent landscape on emerging ordering technologies, and the European foodservice competitive set — all knowable through structured collection.	**Discoveries.** Unanticipated technology paradigms, new entrants, or sudden shifts in customer expectations or regulation that could disrupt the digital ordering space — only surfaceable through systematic scanning.

Implication for scope. The matrix highlights that the project's value lies primarily in (a) converting gaps into open questions (the KIQs), and (b) using structured analysis to surface what is currently tacit or undiscovered. The project should not waste effort restating known assumptions but should commit time to the right side of the matrix.

2. 5Ws & 2Hs — deriving the scope statement

The 5Ws & 2Hs framework progressively narrows from business goal to operational scope. Each question must trace a direct line back to the goal; otherwise the project drifts out of scope. Applied to METRO:

Question	Answer for METRO
Why The business goal	METRO must defend and grow its position as the leading European B2B wholesaler for HoReCa amid digital disruption. Customer retention and competitive positioning in digital channels are decisive for the sCore 2030 targets (40% digital sales).
Who The users / actors	Professional HoReCa customers in Europe (restaurants, hotels, caterers); indirectly, METRO's strategy and digital teams who consume the intelligence output to inform decisions.

How The business impact / mechanism	Digital ordering capabilities act as retention and convenience drivers; competitor evolution and emerging technologies shape what is competitively necessary over a 3–5 year horizon.
What The deliverable / focus	Customer-facing digital services — specifically the e-commerce ordering platform and the mobile commerce app. Internal tools, logistics, and supplier-side software are excluded.
Where Geographical boundary	Europe (METRO's principal HoReCa market).
When Time horizon	A 3–5 year forward horizon, with retrospective context drawn from publicly available data up to the present.
How much Economic / effort constraint	Bounded by course resources: open-source data (SCOPUS, Espacenet), no paid data, no web scraping, R-based analysis only.

The convergent reading of these seven answers yields the scope statement:

> *"This project examines METRO AG's customer-facing digital services for professional HoReCa customers in Europe, with a focus on its e-commerce ordering platform and mobile commerce app, in order to assess how these capabilities contribute to customer retention and competitive positioning over a 3–5 year horizon."*

3. Centering the scope — Generality, Consistency, Focus

The lecture highlights three centering criteria, each deliberately balanced:

Criterion	Application to METRO scope
Generality vs Specificity	Too broad would have been *"digital transformation of METRO."* Too narrow would have been *"the reorder button in the German mobile app."* The chosen scope — customer-facing digital ordering channels in European HoReCa — is operational: specific enough to guide collection, broad enough to span KITs.
Consistency	Every element of the scope is internally coherent: *customer-facing* aligns with *customer retention*; *HoReCa* aligns with the sCore strategic segment; *3–5 year horizon* aligns with sCore's 2030 targets. No element pulls in a contradictory direction.
Focus	Several adjacent topics — internal logistics, supplier-side software, credit/invoicing, the DISH hospitality tools — are deliberately excluded so that the analysis stays centered on retention and positioning rather than diffusing across METRO's full operational footprint.

4. Negative Analysis — defining what is out of scope

Negative analysis sharpens the scope boundary by stating explicitly what is excluded and why. Each exclusion is a deliberate choice, not an oversight:

Excluded element	Reason for exclusion
Internal digital tools (CRM, ERP, employee-facing systems)	Not customer-facing; do not directly affect customer retention or perception.

Logistics, transport, and warehouse systems	Back-office infrastructure; relevant to operational efficiency but outside the customer-experience focus of this project.
Supplier-side software and procurement-side platforms	Upstream, not downstream; outside the HoReCa-customer relationship.
Credit management, invoicing, billing workflows	Financial back-office processes; not part of the ordering experience.
Non-European markets (METRO Asia)	Different competitive and regulatory contexts; would dilute focus.
The Real retail / hypermarket legacy	Divested business; not part of METRO's current strategic identity.

Note on geographic exclusion. The exclusion of non-European markets refers to METRO's market and competitive context as the *object* of analysis. Evidence whose *origin* happens to be outside Europe (international patent databases, global academic literature) remains admissible — the relevant distinction is between the geography being studied and the geography of the source.

5. From Scope to KITs — Clarity, Specificity, Usefulness

The lecture identifies three KIT categories (strategic decisions, early warning, key players) and three articulation criteria (clarity, specificity, usefulness). The three KITs defined for METRO map cleanly to the three categories:

KIT	Category	Why it satisfies Clarity / Specificity / Usefulness
KIT 1. Effectiveness of digital ordering channels in customer retention and convenience.	Strategic decisions & actions *(Are we flying blind?)*	Directly tied to sCore execution; specific to digital ordering capabilities METRO has already deployed; useful because it informs ongoing investment decisions in those channels.
KIT 2. Emerging changes in customer expectations, competitor strategies, and digital service trends.	Early warning *(Are we looking ahead?)*	Forward-looking by construction (3–5 year horizon); specific to three dimensions (expectations / competitors / trends); useful because it surfaces threats and opportunities before they materialize.
KIT 3. Competitive positioning of METRO's e-commerce platform and mobile app relative to traditional and digital-native players.	Key players & positioning *(What is our position?)*	Frames METRO against named competitor categories; specific to the customer-facing digital layer; useful because it grounds positioning claims in evidence rather than assumption.

The three KITs form a **dynamic system**: strategic decisions in KIT 1 are informed by early warnings in KIT 2 and constrained by positioning in KIT 3. None stands alone.

6. KIQ formulation — SMART and question types

KIQs translate KITs into answerable questions. The lecture material specifies the SMART framework (Specific, Measurable, Achievable, Relevant, Timely) and three question types (Polar, Alternative, Q-word). The KIQs in this project follow these principles:

Principle	How applied in this project
SMART — Specific	Each KIQ names a concrete object of analysis (functionalities, retention signals, emerging trends, competitor capabilities, customer perception).
SMART — Measurable	Each KIQ is paired with explicit indicators (patent filings, scientific publications, app-store reviews, feature comparisons).
SMART — Achievable	All KIQs are answerable through course-permitted data sources (SCOPUS, Espacenet, public web content); none requires paid databases or web scraping.
SMART — Relevant	Every KIQ traces back to the scope statement and to METRO's sCore 2030 targets around digital sales, retention, and competitive positioning.
SMART — Timely	Each KIQ is bounded by the 3–5 year horizon defined in the scope, with the analytical lens calibrated for recent data (typically 2018–2026).
Question types	KIQs are predominantly **Q-word** (What / How / Who) — the standard form for exploratory intelligence questions. Polar and Alternative forms are available as derived sub-questions during analysis.

7. Indicators — dimensions and types of analysis

Indicators provide observable evidence to answer KIQs. The lecture identifies three dimensions (Time, Geographic, Semantic) and six analysis types. This project operates principally on two dimensions and two analysis types — a deliberate scoping choice given course constraints:

Element	Use in this project
Time dimension	Trend curves of patent filings and academic publications over recent years reveal whether signals are accelerating, stable, or fading.
Semantic dimension	Text-mining techniques (occurrence, co-occurrence, n-grams) align findings with the language of digital ordering and retention discourse.
Geographic dimension	Where corpus data permit, geographic concentration of patent activity and applicant origin is examined, to surface where innovation in digital ordering is taking place.
Descriptive analysis	Frequency counts of keywords, IPC codes, and applicants — to establish a factual baseline.
Exploratory analysis	Co-occurrence networks and bigram analysis — to uncover patterns and relationships between concepts (hypothesis-generating).
Inferential / Predictive / Causal / Mechanistic	Not applied. These would require statistical sampling, formal hypothesis testing, or controlled experiments that exceed the course scope and the proxy-based data strategy.

Summary. The METRO scope, KITs, and KIQs result from a structured derivation: Rumsfeld surfaces what needs to be known; 5Ws & 2Hs converges on the scope statement; centering criteria balance generality and focus; negative analysis sharpens the boundary; KIT categories and KIQ formulation principles operationalize the scope into the Intelligence Cycle. *Reference: SCI — Scope, KITs and KIQs, I. Spada, University of Pisa, A.Y. 2025/2026.*